King Saud University
College of Computer and Information Sciences
Department of Information Technology
Semester-2 1447H

# Eyecept Report

**Prepared by**
Shahad Alotaibi, 444201168
Lana AlHumaidan, 444200238
Fatema AlHashem, 444203021
Joud Aljamaan, 444200201

**Supervised by**
Dr.Ameera Almasoud

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CHAPTER.1

# Introduction

# 1 Introduction

Security in high-value environments such as museums and jewelry exhibitions extends beyond merely monitoring physical spaces. It encompasses protecting valuable assets, preserving cultural heritage, and ensuring the safety of visitors. Since valuable items are often openly displayed and relatively accessible, these locations naturally become potential targets for theft attempts. Despite the widespread deployment of modern surveillance cameras, most existing surveillance systems primarily focus on recording incidents after they occur, rather than understanding behavioral patterns or enabling early intervention when risk indicators begin to emerge.

In practice, theft rarely occurs suddenly. It is typically preceded by subtle behavioral signals that may reflect emerging intent before an actual incident takes place. Although these behaviors may appear ordinary or insignificant at first glance, analyzing them within their temporal and environmental context may reveal a gradual escalation in risk levels.

In crowded environments, security personnel are required to monitor multiple screens simultaneously, creating continuous cognitive pressure and increasing the likelihood that early warning signals may be overlooked. As this operational burden accumulates, relying solely on human observation becomes insufficient for detecting risks at their earliest stages. In high-value settings, delayed detection may lead to financial losses and institutional reputational damage, making early identification of risk indicators critically important.

Furthermore, many current intelligent surveillance solutions remain reactive in nature, generating alerts only after a clear violation has already occurred. This limitation highlights the need for more advanced systems capable of supporting real-time behavioral analysis, reducing false alarms, and assisting security teams in prioritizing risks more effectively.

To address these challenges, this project proposes an intelligent surveillance system named Eyecept. The system is designed to support security teams in detecting early risk indicators before they escalate into actual threats. It aims to transform surveillance from a reactive process focused solely on post-incident documentation into a proactive analytical tool that enhances operational efficiency and informed decision-making. This report presents the identified problem, the proposed solution, the project objectives and scope, and an overview of the system architecture.

## 1.1 The Problem

Surveillance systems are widely used in high-value environments such as museums and jewelry exhibitions to monitor visitor movement and document events as they occur. While these systems provide essential support for observation and post-incident review, they remain limited in their ability to interpret subtle behavioral signals that may indicate emerging risk before an actual incident takes place.

In many real-world situations, security incidents do not happen suddenly. Instead, they are often preceded by minor behavioral cues that may appear ordinary or insignificant at first glance, especially in dynamic and crowded environments. For example, a visitor may repeatedly direct their attention toward a specific valuable item without physically interacting with it. Although such behavior may seem harmless when viewed in isolation, its repetition within a short period of time may reflect a developing risk pattern. However, current surveillance systems lack the analytical depth necessary to reliably distinguish between natural curiosity and behavior that may signal an elevated level of risk.

Moreover, most existing surveillance systems are primarily designed to document clear violations after they occur, reinforcing a reactive rather than preventive operational model. As a result, early behavioral indicators often go unnoticed, particularly when they are subtle, fragmented, or highly context-dependent. This creates a critical temporal gap between the emergence of the first warning signal and the moment effective intervention becomes possible.

Although some commercial solutions incorporate technologies such as motion tracking or pose estimation, significant limitations remain in their ability to interpret behavioral context, assess sustained attention patterns, and support proactive risk identification without generating excessive false alarms. Consequently, a clear operational gap persists in current surveillance practices, particularly in detecting subtle pre-incident behavioral indicators before they escalate into actual threats.

## 1.2 The Solution

In high-value environments such as museums and jewelry exhibitions, most surveillance systems focus on recording incidents after they happen, which limits the ability of security teams to respond early. Subtle behaviors that often appear before a theft attempt are usually overlooked, especially in crowded areas.

For example, a visitor may stand in front of a valuable item, keep their head and gaze directed toward it for an extended period, and slowly move their hand closer to the display. Since no clear violation has occurred, traditional systems do not flag this behavior, even though it may indicate a potential theft attempt.

The Eyecept system addresses this issue by focusing on a small set of reliable pre-theft indicators: hand movement, head orientation, and attention duration. These behaviors are analyzed locally using computer vision techniques, without relying on cloud processing. When unusual patterns are detected, the system sends a real-time alert to security staff, enabling early intervention while maintaining visitor privacy.

## 1.3 Product

In high-value environments such as museums and jewelry exhibitions, most surveillance systems focus on recording incidents after they happen, which limits the ability of security teams to respond early. Subtle behaviors that often appear before a theft attempt are usually overlooked, especially.

### 1.3.1 Product Vision

| |
|---|
| Product Vision:<br><br>For a museum and jewelry exhibition security teams<br><br>Who require early detection of suspicious behavior<br><br>The Eyecept is a web-based intelligent surveillance system<br><br>That detects pre-theft behaviors in real time<br><br>Unlike traditional CCTV systems like Avigilon Video Analytics<br><br>Our product analyzes subtle behaviors locally while preserving privacy |

## 1.3.2 Product Roadmap

Figure 1- displays the roadmap for the Eyecept platform.

*Figure 1: The Eyecept Platform Roadmap*

## 1.3.3 Objectives

### Product (customer focus-value):

Eyecept is a smart surveillance system that is used in high-value locations like museums and jewelry shows. The system facilitates early identification of suspicious behavior through examination of behavioral pointers in real time. It helps security guards by giving them instant alerts but all the processing of video is done locally to maintain privacy and minimize time-to-response.

**The product objectives of Eyecept are as follows:**

1. **Enable real-time hand movement detection:** Identify and detect hand movements around valuable items to detect suspicious activities before physical contact.
2. **Analyze head orientation and attention duration**: Track head pose and the length of time an individual focuses on a specific item to infer sustained interest that may indicate pre-theft intent.
3. **Detect protected-zone intrusions:** Establish sensitive spatial areas around exhibits of high value and raise alerts in case hands or bodies intrude into these spaces too soon.

4. **Provide instant suspicion notifications to security guards:** Send real-time notifications in case suspicious behavior patterns are identified to facilitate more efficient and quicker human reaction.
5. **Ensure on-device, offline video processing:** Perform all analysis locally on edge devices to preserve privacy, reduce latency, and eliminate dependence on cloud connectivity.
6. **Support multi-camera suspect tracking:** Keep constant surveillance of a target of interest from several camera perspectives within the facility.
7. **Documented proof in the form of an automatic snapshot or short incident video:** Record and hold timestamped image evidence on the local machine to be reviewed, reported, and analyzed after the fact.

- **Project (solution focus-plan):**

**The Eyecept project will require the following steps to be completed:**

1. **Requirements analysis and use-case definition:** Find out the security requirements in museums and jewels exhibitions, pre-theft behavior conditions, secure areas, and alarm states.
2. **Selection and preprocessing of online datasets:** It concerns the selection of appropriate publicly available surveillance databases, screening of the pertinent situations, and an initial processing of video data (frame extraction, labeling refinement, and quality correction) to fit the specific environment.
3. **Model selection and integration:** Pre-trained models of hand detection, head pose and person tracking need to be integrated and be trained on CCTV-like video.
4. **Development of behavior analysis logic:** Develop logic to analyze hand movement, head orientation ,and attention duration ,and use this information to determine early suspicious behavior.
5. **Implementation of zone and intrusion detection:** Design value-based areas of interest (ROIs) around valuable objects and identify the intrusion before physical contact.
6. **Evidence and alerting system:** Introduce real-time notifications to the security guards and automatically record snapshots or short clips with timestamps.
7. **Edge deployment and performance optimization:** Deploy the system on an edge device and optimize performance for offline, on-device processing.
8. **Monitoring dashboard development:** Develop a simple web-based dashboard to visualize alerts, events, and recorded incidents.
9. **Testing, evaluation and refinement:** Test the system with a variety of cases ,and camera views, determine the accuracy and latency, and make changes to the system.
10. **Documentation and final integration:** Final preparation and documentation of the system, and preparation for the final demonstration and evaluation.

- **Learning (student focus):**

1. **Computer vision on surveillance systems**: Practical expertise in integrating and deploying statistical models of computer vision like YOLO to detect objects, and hands, Hopenet to estimate the head pose, and ByteTrack to track people in actual CCTV applications.
2. **AI system design based on behavior:** Developing the ability to transform raw model outputs into useful behavioral attributes through integration of a series of indicators (hand movement, attention duration and spatial context) to identify the early signs of suspicious behavior.
3. **Edge AI and on-Device processing:** Practical experience deploying AI models on edge devices, optimizing performance for real-time, offline processing, and understanding the trade-offs between cloud-based and local computation.
4. **Multi-camera coordination and tracking: The** knowledge of how several cameras can team up and follow a target subject in various positions and still have the same identity throughout an area under surveillance.
5. **Alerts and notifications in real-time:** How to create and establish event-driven alerts and push notifications to alert the security staff immediately in case suspicious activity has been detected.
6. **Local event logging and storage of data:** Capability to achieve experience with the local storage of snapshots, incident clips, and alert metadata, as well as data flow between the processing of AI, storage, and the monitoring dashboard.
7. **Agile system integration and planning:** Training to break down a complex system into manageable modules, scheduling work into sprints, managing progress on a task management tool, and incorporating of multiple components into a working system.

### 1.3.4 Scope

The Eyecept project aims to enhance security in museums and jewelry exhibitions by detecting early suspicious behavior using video analysis. The system will process video locally from USB or IP cameras and provide real-time alerts to security personnel. It will initially support the English language and operate as an academic prototype in controlled environments. The project focuses on behavioral detection (head orientation, hand movement, and protected-zone intrusion) and does not include identity recognition or large-scale commercial deployment. The scope may be refined as development progresses.

### 1.3.5 Hardware/Software Tools and Cost

Table 1- displays the hardware/software tools and cost for the Eyecept platform.

*Table 1:The Eyecept Platform Hardware/Software Tools and Cost Overview*

| Hardware Tools | |
|---|---|
| Name and Description | Cost<br>*mention date it will be available* |
| Laptop for coding | No cost |
| USP or IP camera | 250–500 SAR |
| Raspberry Pi 5 / Jetson Nano for edge processing | 800–1500 SAR |
| External SSD storage | 400-700 SAR |
| Stable network connection for alerts | 200-600 SAR |
| **Software Tools** | |
| Name and Description | Cost |
| Word for documentation | No cost |
| Jupyter Notebook or google Collab | No cost |
| Python 3.10+ | No cost |
| PyTorch | No cost |
| OpenCV | No cost |
| Ultralytics YOLO | No cost |
| ByteTrack or Norfair (tracking) | No cost |
| Hopenet (head pose estimation) | No cost |
| NumPy / Pandas | No cost |
| Roboflow for video processing | No cost |
| Average Total Cost | 2475 SAR |

## 1.4 Scrum Team

### 1.4.1 Skill Set Requirements

Table 2- displays the key skills requirement for the team of the Eyecept platform.

*Table 2:The Eyecept Platform's Team Key Skill Requirements.*

| Technical Skill Required | What is the current level of the team (*beginner-intermediate- advanced*) for each skill? How will the gap be bridged? *(if necessary*) Learning plan |
|---|---|
| Python Programming Language | Intermediate |
| Computer Vision & Deep Learning | beginner |
| Video Processing and Image Analysis | beginner |
| Edge-Based AI Deployment | beginner |
| Database Design & Local Storage | Intermediate |
| Web-Based Monitoring Interface Development | Advanced |

### 1.4.2 Learning

In the planning stage of the Eyecept project, the team developed a better ability to break down complicated real-world issues and express them into organized system elements. The team also learnt how to break down a massive intelligent surveillance system into manageable subdivisions such as detection, tracking, behavioral analysis, alert generation, and dashboard monitoring.

Another practicable experience that the team gained was the utilization of Jira to manage agile projects, including organizing the items of a product backlog, setting priority requirements, breaking tasks into manageable units, meeting minutes, and various project artefacts. As a result, the team was able to come up with a clear understanding of structured work workflows and collaborative project management. Besides this, the project enhanced the research and analysis skills of the team. The team has examined various scholarly articles and commercial systems, contrasted the available solutions, found research gaps, and used the same in streamlining and validating the proposed system design. In addition, the team also acquired a better understanding of the value of edge-based processing in privacy-sensitive settings and how architectural choices have the potential to affect system performance, latency, and data protection directly.

### 1.4.3 Roles and Responsibilities

Table 3- displays the roles and responsibilities for the team of the Eyecept platform

*Table 3:The Eyecept Platform's Team Roles and Responsibilities Overview*

| Scrum Team | |
|---|---|
| Product Owner:<br>(list name of the project supervisor) | Dr. Ameera Almasoud |
| Developers: (list student names) | Fatema AlHashem<br>Joud Aljamaan<br>Shahad Alotaibi<br>Lana AlHumaidan |
| Scrum Master (SM):<br>(list name of designated SM for the project) | Dr. Ameera Almasoud |
| Stakeholders:<br>(list names of project stakeholders, include customer names if any) | Dr. Ameera Almasoud, Examination team |

CHAPTER.2

# BACKGROUND AND LITERATURE REVIEW

# 2 Background

**• Computer Vision in Surveillance Systems**

Computer Vision is a field of Artificial Intelligence that enables machines to interpret and analyze visual data such as images and video streams. In modern surveillance systems, computer vision techniques are used to detect objects, track individuals, and analyze behaviors in real time. In the Eyecept system, computer vision is essential for detecting hand movement, analyzing head orientation, and monitoring suspicious behavior before theft occurs.

**• Object and Hand Detection**

Object detection is a technique used to identify and locate objects within video frames. In surveillance systems, it is used to detect people and specific body parts such as hands. Eyecept relies on hand detection to monitor slow or unusual movements around valuable objects, which may indicate potential pre-theft behavior.

**• Head Pose Estimation and Attention Analysis**

Head pose estimation determines the orientation of a person's head to understand where they are looking. In high-value environments, prolonged attention toward a specific object may indicate suspicious intent. Eyecept analyzes head orientation and attention duration to identify abnormal focus patterns.

**• Region of Interest (ROI)**

A Region of Interest (ROI) is a defined area within a video frame that receives focused monitoring. In Eyecept, protected zones are created around valuable objects. Any movement inside these zones is analyzed to determine whether it represents suspicious behavior.

**• Edge Computing**

Edge computing refers to processing data locally on a device rather than sending it to the cloud. Eyecept performs all video processing on an edge device to reduce latency, ensure faster response, and preserve visitor privacy.

# 3 Literature Review

The rapid development of Artificial Intelligence and Computer Vision technologies has significantly transformed modern video surveillance systems. Traditional surveillance solutions mainly focus on monitoring and recording events after they occur, which limits their ability to prevent security incidents before they happen. Recently, research has shifted toward intelligent video analytics systems capable of detecting suspicious behaviors in real time and supporting proactive security decision-making [1].

In high-value environments such as museums and jewelry exhibitions, early detection of suspicious behavior is critical. Studies in behavioral analysis suggest that subtle indicators such as prolonged attention toward valuable objects, repeated head orientation, and abnormal hand movements near secured items can indicate potential malicious intent before an actual theft attempt occurs. Real-world security incidents in high-value environments demonstrate how quickly theft events can occur and highlight the need for intelligent preventive monitoring systems [2]. However, many existing surveillance solutions are designed to detect obvious events such as motion, intrusion, or object removal rather than subtle pre-incident behavioral patterns.

Several intelligent surveillance and video analytics solutions currently exist in the global market. For example, Veesion provides AI-based video analytics designed to detect shoplifting gestures and suspicious hand movements in retail environments. Similarly, Vaidio is an AI video analytics platform widely used in smart cities and enterprise surveillance, offering features such as abnormal behavior detection, intrusion detection, and object recognition. Additionally, companies such as Hikvision, Dahua, Bosch, and Axis Communications provide enterprise-level surveillance systems focusing on motion detection, object tracking, and intrusion alerts as part of general security monitoring solutions.

Furthermore, biometric and behavior analytics companies such as 3DiVi focus primarily on identity recognition technologies, including face recognition and body tracking, which are mainly used for access control and identification rather than behavioral prediction. Regional providers such as ACIX Middle East offer integrated physical security solutions focusing on operational monitoring, crowd movement tracking, and post-incident analysis.

Despite the strong capabilities of these systems, most of them follow a reactive security model that focuses on detecting events after they occur. In addition, many solutions rely heavily on cloud-based processing, which may introduce latency and raise privacy concerns, especially in sensitive environments such as museums and exhibitions. Edge computing has emerged as an important alternative approach, enabling video data to be processed locally on the device, which reduces latency, improves response time, and enhances data privacy [3].

Recent research trends highlight the importance of edge computing in surveillance systems. Edge-based processing enables video data to be processed locally on the device, reducing latency, improving response time, and enhancing data privacy. This approach is particularly important in environments where data confidentiality is critical [3].

Based on the identified limitations in existing systems, the Eyecept system is designed to provide a proactive behavioral security solution that focuses on detecting pre-theft behaviors using multi-factor behavioral analysis. The system combines real-time hand movement detection, head orientation analysis, and attention duration monitoring to identify suspicious behavioral patterns around high-value objects.

In addition, Eyecept supports protected zone monitoring using Region of Interest (ROI) analysis and generates instant alert notifications when suspicious behavior is detected. The system also supports multi-camera suspect tracking, automatic short-clip recording upon detection of suspicious activity and provides a centralized monitoring dashboard for real-time security monitoring and decision-making.

Eyecept is also designed using on-device edge processing, which enhances privacy protection and reduces dependence on cloud infrastructure. This design makes the system suitable for deployment in privacy-sensitive environments such as museums, jewelry exhibitions, and high-security facilities [3].

Moreover, Eyecept is particularly suitable for deployment in regions such as the Middle East and Saudi Arabia, where there is an increasing demand for intelligent security solutions that ensure privacy preservation while maintaining high detection accuracy and rapid response capabilities.

Overall, the literature review demonstrates the need for proactive behavioral security systems that go beyond traditional surveillance monitoring. The analysis of existing solutions highlights the research gap in detecting subtle pre-incident behaviors. Eyecept aims to address this gap by providing an intelligent, privacy-preserving, and real-time behavioral surveillance solution specifically designed for high-value asset protection environments.

## 3.1 Competitive Product Analysis

**1-Veesion**

*Figure 2-Veesion Logo*

The logo shown above belongs to **Veesion**, a French company that provides an AI-based video analytics system designed to detect shoplifting gestures and suspicious hand movements in retail environments. The system focuses on recognizing theft-related gestures and sending real-time alerts to security personnel.

Unlike Eyecept, which is deployed in museums and jewelry exhibitions and targets pre-theft behavior using subtle indicators such as prolonged attention and gradual hand movement, Veesion mainly detects theft during or immediately before the act. Moreover, Veesion relies on cloud-based processing, while Eyecept operates entirely on edge devices to ensure privacy preservation and on-site real-time response. [4]

**2-Vaidio**

*Figure 3-Vaidio Logo*

The logo shown above belongs to **Vaidio**, an AI-powered video analytics platform developed by IronYun and widely used for security surveillance, smart cities, and transportation monitoring. It provides features such as abnormal behavior detection, intrusion detection, and object and person recognition.

In contrast to Eyecept, which focuses specifically on detecting subtle pre-theft behaviors in high-value environments such as museums and jewelry exhibitions, Vaidio is a general-purpose surveillance system that analyzes overall movement and anomalies in the scene rather than fine-grained attention and hand movement patterns. Furthermore, Eyecept is designed for privacy-preserving edge deployment, while Vaidio often relies on centralized or cloud-based architectures. [5]

**3-DiVi**

Figure 4-3DiVi Logo

The logo above belongs to **3DiVi**, a computer vision and biometric technology company that develops systems for face recognition, body tracking, and behavior analysis. Its solutions are used in security and access control applications to identify individuals and analyze their movements.

Unlike Eyecept, which aims to predict pre-theft behavior using subtle cues such as head orientation and attention duration toward valuable objects, 3DiVi mainly focuses on identity recognition and general behavior analytics. Additionally, Eyecept is designed explicitly for privacy-sensitive environments and operates locally on edge devices, reducing privacy risks associated with biometric identification systems. [6]

4- **Hikvision**

Figure 5-HikiVision Logo

The logo above belongs to **Hikvision**, a Chinese company specializing in large-scale CCTV implementations, focusing on person and vehicle detection, motion analytics, and intrusion alerts as part of comprehensive security surveillance solutions.

Eyecept is unique in its setting and purpose, as it is deployed in museums and jewelry exhibitions, to prevent pre-theft behaviors. Instead of relying on general movement in the scene, it utilizes subtle indicators (e.g., prolonged attention and hand movements in secured areas). Since Eyecept operates entirely locally on edge devices, it can react on-site and maintain data confidentiality. [7]

5- **Dahua Technology**

Figure 6-Dahua Technology Logo

The logo above belongs to **Dahua Technology**, a Chinese company that offers surveillance systems primarily designed to monitor visible activity through motion and person detection, and as well as broad behavior analytics, sometimes with cloud storage capabilities.

Eyecept shifts the analysis from scene-level activity to object-level risk by computing hand movement within item-associated ROIs and directly activating real-time audio cues and instant notifications. It also has the advantage of being lightweight and edge-based, making it easier to deploy while enhancing privacy in sensitive settings. [8]

## 6-**Bosch**

*Figure 7-Bosch Logo*

The logo above belongs to **Bosch**, a German company that offers sophisticated video analytics capable of generating alerts when motion or suspicious activity is detected. The data can be stored either on premises or in the cloud, with a focus on actions occurring within the environment.

Eyecept, in contrast, was created specifically for museums and jewelry exhibitions and is oriented toward recognizing hand movement before or during a theft attempt. Unlike Bosch systems, Eyecept implements a threat-level-based alerting model, where alert responses increase in strength, ranging from monitoring to dashboard alerts or direct security warnings. [9]

## 7-Axis Communications

*Figure 8-Axis Communications Logo*

The logo shown above belongs **to Axis Communications**, a Swedish company and a global leader in network video surveillance solutions. It provides high-quality IP cameras and video analytics used in airports, smart cities, and large enterprises.

Axis systems mainly focus on monitoring and recording events after they occur, with limited capabilities in detecting subtle pre-theft behaviors. Additionally, Axis solutions are generally considered high-cost compared to lightweight edge-based systems such as Eyecept. [10]

8-**Hanwha Vision**

*Figure 9-Hanwha Vision Logo*

The logo shown above belongs to **Hanwha Vision**, a South Korean company specializing in AI-powered surveillance solutions. It offers smart cameras and video analytics for motion detection and object classification in commercial and public facilities.

Despite incorporating AI technologies, its solutions mainly focus on general surveillance tasks and do not specifically analyze detailed pre-theft behavioral indicators such as head orientation or attention duration toward valuable objects. [11]

9-**ACIX Middle East**

The logo shown above belongs to **ACIX Middle East**, a regional provider of integrated physical security and surveillance solutions, offering video analytics as part of comprehensive security systems.

*Figure 10-ACIX Middle East Logo*

Its analytics primarily focus on detecting observable events within monitored environments, such as unauthorized access, zone intrusions, crowd movements, and general activity tracking. These solutions are mainly intended to support operational security and post-incident investigation rather than interpreting subtle behavioral cues. [12]

## 3.2 Comparison for competitor Systems

The competition analysis table that would be compared with the Eyecept would be the comparative analysis of Eyecept with some of the most successful surveillance and video analytics systems. The main functional and architectural features that can be taken to the high value environment such as museums and jewelry shows are juxtaposed. All these are hand gesture recognition, focus and head location, security zone monitoring, real-time alarm, screen edges, multi-camera tracking, edge detection and dashboard, automatic evidence capture and dashboard detection.

*Table 4:Comparison of competitors*

| Features / competitors | Real-time hand movement detection | Head orientation & attention duration analysis | Protected-zone (ROI) intrusion alert | Instant suspicion notifications | On-device, offline edge processing | Multi-camera suspect tracking | Automatic snapshot / short-clip recording | Monitoring dashboard | Solution Types |
|---|---|---|---|---|---|---|---|---|---|
| Veesion | ✓ | | | ✓ | | | ✓ | ✓ | Retail video Analytics |
| Vaidio | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | AI vision platform |
| 3Divi | | | | ✓ | ✓ | ✓ | ✓ | ✓ | Biometric + CV analytics |
| Hikvision | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Enterprise CCTV + analytics |
| Dahua Technology | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Enterprise CCTV + analytics |
| Bosch | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Video analytics |
| Axis Communication | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Network video |
| Hanwha Vision | | | ✓ | ✓ | | ✓ | ✓ | ✓ | AI survellance |
| Acix middle east | | | ✓ | ✓ | | ✓ | ✓ | ✓ | Security solutions provider |
| Eyecept | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | Web monitoring |

As indicated in the comparison in **Table 4**, the competitive advantages are those unique to the Eyecept system when compared to a number of well-known video analytics systems and intelligent surveillance systems in the world like behavioral analysis systems, biometric analytics systems, and the traditional enterprise surveillance systems. In contrast to the comparison, the fact that the Eyecept can be used to facilitate proactive security activities in high value environments like museums and jewelry shows depicts that the capability can be employed to prevent security incidents that would otherwise take a long way before security event would happen. Most of the available surveillance systems have powerful features like the motion detection, object tracking and recording of occurrences but are usually configured to give reactive response to the surveillance or analyze a situation which has already occurred. Eyecept, on the other hand, has pro-active security model, which is established on a multi-factor behavioral analysis. The system has real time hand movement recognition, head orientation recognition, and head attention time recognition so as to pick the tiniest behavioral clues of the threat of theft that is not always detectable by the conventional surveillance solutions. Furthermore, Eyecept offers

the possibility to track the secure zones via the analysis of the Region of Interest (ROI) and an immediate notification was made in case of the deviant or suspicious behavior. This will allow the security personnel to respond quickly hence averting an attack before security breakage. Moreover, Eyecept will be compatible with on-device edge processing that can improve the confidentiality of the information, does not require the utilization of the cloud infrastructure, and limits the system latency. The system also allows the multi-camera tracking of the culprits where upon the detection of suspicious activities, automatic short clip recording is done and also allows central control monitoring dashboard that can be used by the security team to make real-time decisions relating to security matters. This type of architecture renders Eyecept very well adapted in such a setup with a high privacy sensitivity rate as well as in places like the Middle East and Saudi Arabia where an increasing demand of smart security systems exists in ensuring the safety of privacy and at the same time, high detection rates and speed. This leads to the fact that Eyecept has a good potential of being practically implemented and working successfully in a real-life environment with high security.

# 4 System Description

This section introduces the Eyecept system by identifying its primary users and outlining their roles and experience within security monitoring environments. It then defines the users' needs based on insights gathered from interviews and questionnaires. Furthermore, the system architecture is presented to provide a clear overview of the main components and explain how they function together. Finally, a Use Case Diagram is included to illustrate how users interact with the system in practical scenarios.

## 4.1 Users

Security Operators describe people who normally work in a surveillance and security monitoring area like an exhibition, or a commercial place, and they are usually aged 23 years and above. They usually have a diploma or a bachelors degree and experience in the running of monitoring systems and responding to security incidents. They have medium technical skills and are not necessarily advanced in artificial intelligence knowledge; however, they are accustomed to working with web-based platforms and control dashboards.

Administrators are experts who compile and arrange the system of the organization. They are usually above 27 years old and have at least a bachelor degree in Information Technology,

Computer Science or any other similar course. They possess average to high level technical skills, and are experienced in system administration, networking and database administration.

Field Security officers are working employees whose duty is to physically react to alerts and deal with possible threats on the ground. They are more likely to be above 21 years of age, with a high school education level, and they are professionally trained in threat management and emergency response. They have rudimentary technical expertise, enough to deal with mobile alerts and communication devices.

## 4.2 Requirements Elicitation

In order to have a clear definition of what would be needed in the Eyecept system, we did a requirements elicitation to gain a clearer insight into the actual problems that professionals working in high-sensitivity surveillance face. The aim was to understand the nature of monitoring as it is currently practiced, where existing systems pose challenges, and what enhancements can possibly be made to enable quicker and more effective security decision-making.

There were two methods of data collection. To begin with, semi-structured interviews were carried out in person with security professionals operating in exhibitions and financial monitoring environments. These interviews gave us the opportunity to have structured questions and at the same time provide us with the flexibility to have the participants elaborate on their experiences, daily challenges, and concerns in greater detail. Second, a structured questionnaire was distributed via Google Forms, and 31 replies were gathered. The questionnaire covered workload monitoring, expectations of the reaction time, system usability, the impact of false alarms, and the openness of AI-assisted support. An open-ended question was also included to gather additional qualitative feedback.

The participants had between 9 and 18 years of experience in surveillance-related roles. Their responses reflect the reality of environments where continuous attention is required, and even small delays in identifying suspicious behavior can result in serious consequences.

The insights gathered from both the interviews and the questionnaire directly supported the refinement of the system's functional and non-functional requirements.

(Appendix A :Interviews for full interview transcripts and Appendix B: Questionnaires for detailed questionnaire results.)

### 4.2.1 Interview Analysis:

1. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage rather than during live monitoring?

Participants explained that suspicious behavior is sometimes identified only after reviewing recorded footage rather than during live monitoring. In addition, access to recordings may require administrative approval, which can delay verification and slow down response actions.

In high-value environments, this workflow increases operational risk. These findings support the need for a proactive surveillance system capable of detecting suspicious behavior in real time without relying on post-incident review procedures.

2. Do you believe that relying solely on human observation makes it difficult to consistently detect small or slow movements in busy exhibition environments?

Interviewees highlighted that relying exclusively on human attention makes it difficult to consistently detect subtle or slow suspicious behaviors, particularly in crowded exhibition environments or during peak hours. They also described the mental demands of monitoring multiple camera feeds for extended periods, noting that sustained concentration increases the likelihood of fatigue and oversight.

This reinforces the need for intelligent behavioral analysis to support operators and reduce the cognitive burden associated with continuous monitoring.

3. How difficult is it to track a person of interest when they move between different camera views or between multiple exhibition areas?

Tracking a person of interest across multiple camera views was described as time-consuming and mentally demanding. Participants indicated that manually following individuals between areas or camera feeds increases workload and reduces situational clarity, especially in busy environments.

This finding strengthens the case for automated multi-camera tracking to improve continuity, reduce operator effort, and support faster decision-making.

4. What factors would make you trust and rely on an AI-based surveillance system in your daily work?

Participants emphasized that trust depends primarily on accuracy, reliability, and the clarity of alerts. They stated that frequent false alarms reduce confidence in the system, whereas alerts supported by visual evidence enhance trust and facilitate faster validation.

These insights highlight that reliability, low false-positive rates, and transparent alert explanations are essential for user acceptance of AI-based systems.

5. During peak hours, how manageable is it for you to monitor all camera screens while maintaining the same level of attention continuously?

Interviewers explained that maintaining consistent attention during peak hours is challenging due to the number of screens and the density of visitor movement. Extended monitoring periods increase mental fatigue, which raises the probability of missing important events.

This supports the necessity of automated assistance to maintain consistent surveillance performance during high-demand periods.

6. When a suspicious situation is identified, what are the usual steps taken by the security team, and where do delays typically occur?

Participants described a multi-step process involving footage review, administrative approval, and coordination with field officers. Delays most commonly occur during verification and approval stages.

Such procedural delays create operational gaps in high-value environments, further emphasizing the importance of real-time detection and immediate alerting mechanisms.

7. What type of alert would be most effective for you during work (visual dashboard alert, sound notification, mobile notification, or a combination of these)?

Participants expressed a preference for immediate and noticeable alerts, ideally combining visual and sound notifications. They also stressed that alerts should include contextual information explaining the reason for activation.

This indicates that multi-modal alerting supported by contextual explanation enhances operational responsiveness.

#### 1.1.1.1 Questionnaire Analysis:

Q1. How many hours per shift do you spend monitoring cameras?

A substantial proportion of respondents reported extended monitoring durations per shift. Specifically, 45.2% reported monitoring for 5–8 hours, 22.6% for 2–4 hours, 16.1% for less than 2 hours, and 16.1% for more than 8 hours.

These findings demonstrate that surveillance tasks require prolonged attention, reinforcing the presence of sustained cognitive load in daily operations.

Q2. During monitoring, how high is your mental workload or focus level required?

The results indicate that monitoring demands significant mental effort. Specifically, 43.8% reported high workload, 40.6% reported medium workload, and 15.6% reported very high workload.

This confirms that surveillance roles involve considerable cognitive pressure, increasing the risk of fatigue-related errors.

Q3. How often do you feel that important events might be missed during live monitoring?

More than half of respondents (51.6%) reported that important events are sometimes missed, while 29% indicated that this occurs often.

These findings highlight the limitations of manual monitoring and strengthen the need for automated behavioral detection support.

Q4. How quickly do you usually need to react when suspicious behavior is noticed?

The data reveal strong expectations for rapid response. Specifically, 43.8% require action within seconds, and 37.5% expect response within one minute.

This emphasizes the operational necessity of low-latency detection and immediate alert delivery.

Q5. How clear and easy to use is your current surveillance system interface?

While 51.6% rated their current system interface as neutral in clarity, 22.6% described it as difficult to use.

This suggests that current surveillance interfaces may not fully support rapid and confident decision-making under pressure.

Q6. How important is it for you to receive alerts with visual evidence (snapshot or short clip)?

An overwhelming majority considered visual evidence essential. Specifically, 61.3% rated it as very important, and 32.3% rated it as important.

This strongly validates the requirement for evidence-supported alerts in the proposed system.

Q7. How much do false alarms affect your work efficiency?

False alarms were identified as a significant concern. Specifically, 38.7% reported a moderate impact, while 22.6% reported a significant impact on their efficiency.

These findings indicate that minimizing false positives is critical to maintaining operational efficiency and trust in the system.

Q8. How useful would it be if the system prioritized the most risky behavior automatically?

A strong majority expressed support for automatic risk prioritization. Specifically, 58.1% rated it as very useful, and 41.9% rated it as useful.

This demonstrates clear demand for intelligent threat-level classification.

Q9. How comfortable are you with AI assisting you in monitoring tasks?

Acceptance of AI assistance was generally high. Specifically, 61.3% reported feeling comfortable, and 25.8% reported feeling very comfortable.

This indicates broad readiness among professionals to adopt AI-supported surveillance tools.

Q10. How important is system reliability (working without crashes or delays) during monitoring?

System reliability was rated as very important by 74.2% of respondents.

This confirms that stability, uptime, and consistent performance are non-negotiable requirements in high-sensitivity surveillance environments.

### 4.2.2 Consolidated Key Findings

Based on both qualitative and quantitative findings, the following conclusions were identified:

- Surveillance roles involve sustained cognitive workload and high mental pressure.
- Reactive monitoring and procedural delays create operational vulnerabilities.
- Human-only observation is insufficient for detecting subtle or slow suspicious behaviors.
- Real-time alerts supported by contextual visual evidence are operationally critical.
- False alarms negatively affect efficiency and reduce trust in the system.
- Automatic risk prioritization is strongly supported by users.
- AI-assisted monitoring is widely accepted among professionals.
- System reliability and low latency are essential performance requirements.

These findings provide a structured foundation for defining and refining the functional and non-functional requirements in the Product Backlog, ensuring that the proposed Eyecept system directly addresses real operational challenges in high-value surveillance environments.

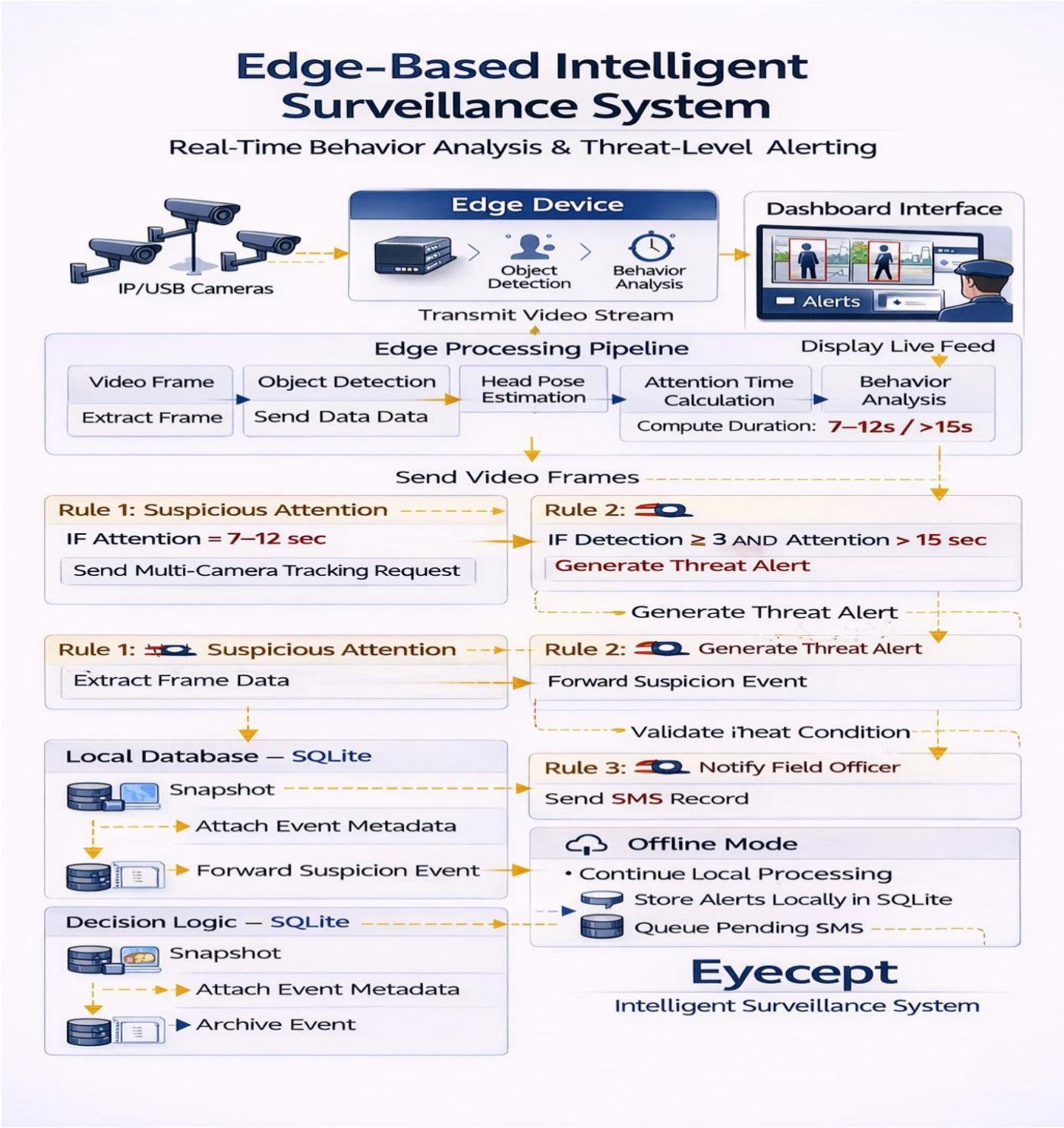
# 5 Architecture

The Eyecept system follows a layered, client–server architecture designed to detect suspicious behavior in real time while preserving privacy. The system is organized into major subsystems: cameras, an edge device, a local SQLite database, a web-based dashboard, and an alert notification system. Live video from USB or IP cameras serves as the input, which is processed locally by the edge device to detect people, track movements, and calculate attention time. Suspicious behavior triggers outputs such as alerts, snapshots stored in the local database, and notifications displayed on the dashboard or sent via SMS to field officers.

The system is divided into three security levels: Level 1 – system administrator managing configuration and database; Level 2 – security operations officer monitoring cameras and the dashboard; and Level 3 – field security officer responding on-site to high-priority alerts. Data is stored locally on SQLite for enhanced security and faster response, avoiding external cloud dependencies.

This high-level architecture ensures that each component has a clear responsibility: cameras provide data, the edge device performs detection and analysis, the database securely logs events, and the dashboard and alert system allow personnel to monitor and respond efficiently. The layered design separates data capture, processing, storage, and user interaction, making the system easier to maintain, extend, and deploy in privacy-sensitive environments.

*Figure 11:System Architecture*



## 5.1 Use Case Diagram

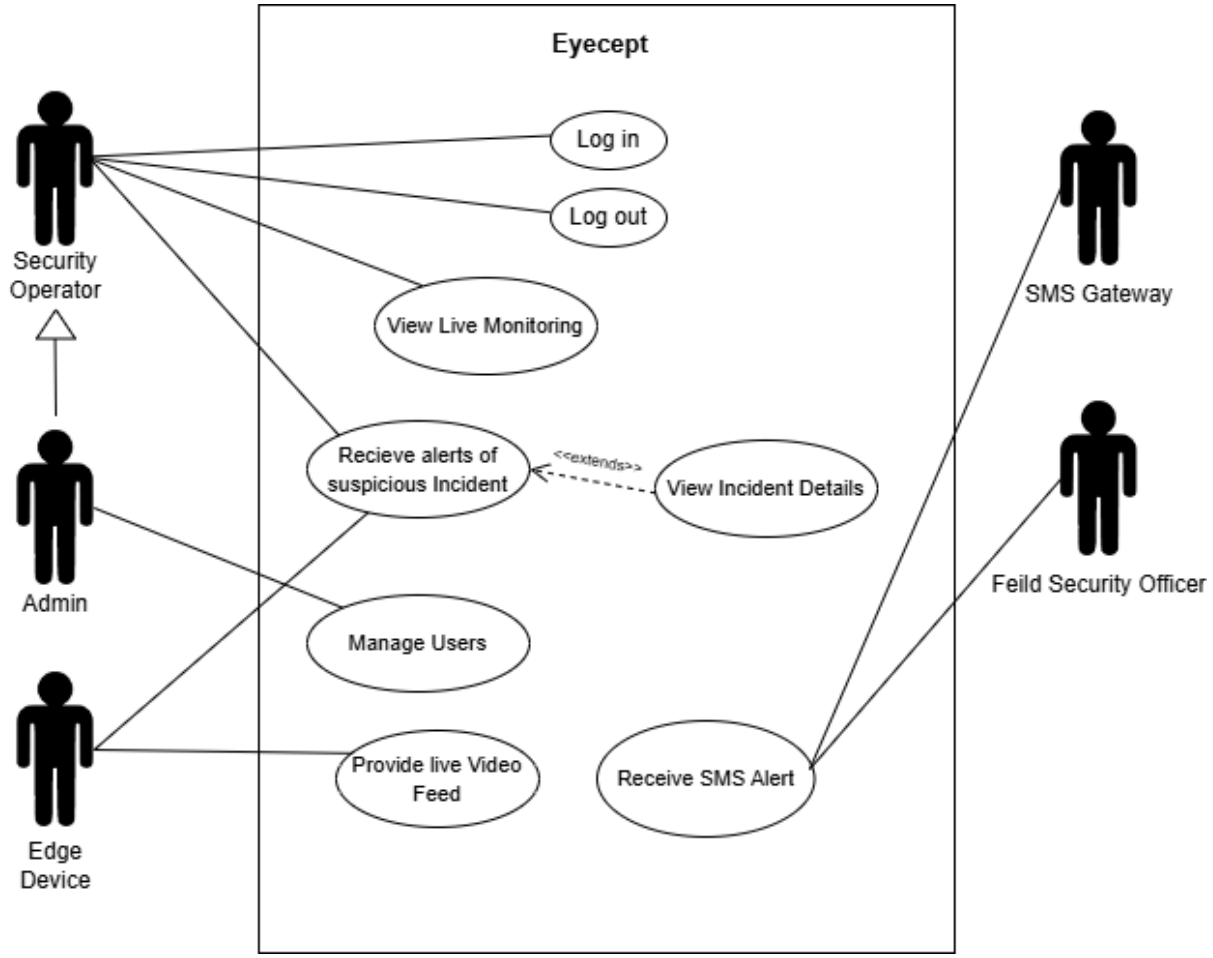



*Figure 12:Use Case Diagram*

# 6 Product Backlog

It is a part which shows the Product Backlog of the Eyecept system. It contains a set of prioritized features and user stories, which are required to deploy the system in small increments. The items in the backlog are referred to the project objectives as well as system requirements and are organized depending on the development priority. In addition to that, the Definition of Ready is provided to have a clear definition of every item in the backlog, which will be technically viable, and ready to be rolled out to the implementation in a sprint.

## 6.1 Definition of Ready

*Table 5: Definition of Ready*

| | Defentions of Ready |
|---|---|
| ✓ | Details are sufficiently understood |
| ✓ | Acceptance criteria are clear and testable |
| ✓ | Estimated and small enough to be completed during sprint |
| ✓ | Dependencies are identified; no blocking dependencies exist |
| ✓ | Performance criteria, if any, are defined and testable |

## 6.2 Product Backlog Table

Table 6 displays the Product Backlog table for the Eyecept, and it contains the user stories for functional and non-functional requirements.

*Table 6:PBI Table*

| PBI (user story) | Size (Story points) | Type (Feature, defect, technical work, knowledge acquisition) | Acceptance Criteria The conditions of satisfaction that must be met for that item to be accepted. |
|---|---|---|---|
| **Functional Requirements** | | | |
| As a security operator, I want to create a new account so that I can access the system. | 5 | Feature | - If the user provides valid required information, then the account must be created successfully.<br>- If required information is missing, then the system must reject the registration request. |
| As a security operator, I want to log in to the system so that I can access the monitoring dashboard. | 3 | Feature | - If valid login credentials are entered, then the system must grant access to the dashboard.<br>- If invalid credentials are entered, then access must be denied. |

| | | | |
|---|---|---|---|
| As a security operator, I want to log out of the system so that I can securely end my session. | 3 | Feature | - If the user selects the log out option, then the session must be terminated. |
| As a security operator, I want to reset my password if I forget it so that I can regain access to my account. | 5 | Feature | - If the user selects “Forgot Password”, then the system must allow password reset.<br>- The user must be able to set a new password successfully. |
| As a security operator, I want to view the monitoring dashboard so that I can monitor detected events. | 5 | Feature | -If the user accesses the dashboard, then live alerts must be visible. |
| As a security operator, I want to receive real-time alerts when suspicious behavior is detected so that I can take preventive action. | 7 | Feature | - If suspicious behavior is detected, then a real-time alert must be sent to security staff. |

| | | | |
|---|---|---|---|
| As a security operator, I want to view incident details so that I can understand detected suspicious behavior. | 6 | Feature | -If the user selects an alert, then the system must display the incident details. |
| As a security operator, I want the system to calculate attention duration so that suspicious behavior can be detected. | 7 | Feature | - If a person remains focused for 7–12 seconds, then the activity must be flagged as suspicious.<br>- If the activity is flagged as suspicious, then the system must instruct other cameras to begin monitoring the same individual. |
| As a security operator, I want to receive a notification alert with a threat level so that I can understand the level of risk.<br>. | 8 | Feature | - If the same person is detected three or more times and attention duration exceeds 15 seconds, then a notification alert must be generated including the threat level. |

| | | | |
|---|---|---|---|
| As a security operator, I want a snapshot to be automatically captured when a threat alert is generated so that the event is documented. | 5 | Feature | - If an alert is generated, then a snapshot must be captured and stored in the event logs including threat level, date, and time. |
| As a security operator, I want to access event logs through the dashboard so that I can review detected incidents. | 5 | Feature | - If an event is logged, then it must be accessible through the dashboard interface. |
| As a security operator, I want to dismiss an alert if it does not require intervention so that false positives are reduced. | 5 | Feature | - If the security determines no intervention is required, then the alert must be dismissible. |

| | | | |
|---|---|---|---|
| As a field security officer, I want to receive an SMS notification immediately when a person's hand enters a predefined ROI so that I can respond immediately. | 8 | Feature | - If a person's hand enters a predefined Region of Interest (ROI), then the system must detect the intrusion.<br>- If ROI intrusion is detected, then an immediate SMS notification must be sent to the field security officer. |
| As an admin, I want to add new user accounts so that authorized staff can access the system. | 4 | Feature | - If the admin enters valid user information, then the account must be created successfully.<br>- If required information is missing, then the system must reject the creation request. |
| As an admin, I want to delete user accounts so that unauthorized users cannot access the system. | 4 | Feature | - If the admin deletes a user account, then that user must not be able to log in. - If a deleted user attempts to log in, then access must be denied. |
| **Non-Functional Requirements** | | | |
| **As** a security operator, **I want** suspicious behavior alerts to appear on the dashboard within 3 seconds of detection **so that** I can act immediately | 5 | Feature | - As a security operator, if suspicious behavior is detected, then the alert must appear on the dashboard within 3 seconds from detection under normal operational load. |
| **As a** security operator, **I want** the system to automatically detect camera disconnections and attempt reconnection within 30 seconds, **so that** monitoring continuity is maintained | 5 | Feature | - As a security operator, if a camera feed is disconnected, then the system must detect the disconnection and initiate a reconnection attempt within 30 seconds. |

| | | | |
|---|---|---|---|
| **As an** admin, **I want** video processing and behavior analysis to run locally with an average inference time of less than 200 milliseconds per frame, **so that** real-time monitoring is maintained without noticeable delay. | 8 | Feature | -As an admin, if video frames are processed on the edge device, then the average inference time must not exceed 200 milliseconds per frame under normal operational load. |
| **As a** security operator, **I want** the monitoring dashboard to be available at least 95% of operational hours, **so that** security monitoring is not interrupted. | 3 | Feature | -As a security operator, if I access the monitoring dashboard during operational hours, then it should be available at least 95% of the time |
| **As a** security operator, **I want** all alerts and recorded snapshots/clips to be stored without any data loss **so that** no critical security evidence is missed | 5 | Feature | -As a security operator, if an alert or a snapshot/clip is generated, then it must be stored with a timestamp and remain accessible without data loss during normal system operation. |
| **As an** admin, **I want** the system to support monitoring of at least 4 cameras simultaneously and operate independently of camera brand**, so that** the system can scale to larger exhibition areas without hardware limitations | 8 | Feature | -As an admin, if I connect up to 4 cameras that support RTSP or ONVIF protocols, then the system must process all feeds simultaneously while maintaining the defined alert latency threshold. |
| **As an** admin, **I want** recorded snapshots and incident clips to preserve visitor privacy through anonymization techniques, **so that** individuals cannot be personally identified from stored evidence. | 8 | Feature | - As an admin, if a snapshot or incident clip is stored, then identifiable facial features must be anonymized, and the data must remain stored locally without transmission to external cloud servers. |

| Enabler Stories | | | |
|---|---|---|---|
| **As a** developer, **I need** to integrate the detection, tracking, and behavior analysis components into a unified processing pipeline **so that** suspicious behavior can be analyzed in real-time. | 8 | Enabler | -As a developer, if a video frame is captured, then the system must process it through detection, tracking, and behavioral analysis sequentially without runtime errors. |
| **As a** developer, **I need** to integrate multiple camera streams(IP/USB) into the system, **so that** real-time video frames can be processed by the AI pipeline. | 5 | Enabler | -As a developer, if a camera is connected, then the system must initialize the stream and capture frames successfully. |
| **As a** developer, **I need** to implement model loading and runtime inference, **so that** the trained AI model can process frames during system operation. | 5 | Enabler | -As a developer, if the system starts, then the trained model weights must load successfully and perform inference on incoming frames. |
| **As a** developer, **I need** to connect the back-end processing engine with the monitoring dashboard **so that** the alert and event logs are displayed to security personnel. | 5 | Enabler | -As a developer, if suspicious behavior is detected, then an alert must be transmitted from the back-end and displayed on the dashboard. |
| **As a** developer, **I need to** prepare and train the YOLO detection model using collected prefetch data sets, **so that** detection accuracy improves for suspicious actions | 8 | Enabler | -As a developer, if the dataset is prepared and used for training, then updated model weights must be generated and validated for improved detection performance. |

| | | | |
|---|---|---|---|
| **As a** developer, **I need** to integrate the trained models onto an edge device, **so that** processing runs locally without cloud dependency | 8 | Enabler | As a developer, if the model is deployed on the selected edge device, then video processing must run locally, and no video data shall be transmitted to external cloud services. |
| **As a** developer, **I need** to integrate a tracking algorithm with the detection model **so that** individuals can be tracked across frames. | 5 | Enabler | As a developer, if the same person appears in consecutive frames, then the system must maintain the same tracking ID under normal movement conditions. |
| **As a** developer, **I need** to implement risk scoring logic based on behavioral indicators **so that** threats of different levels can be classified. | 5 | Enabler | As a developer, if suspicious behavioral indicators are detected, then the system must calculate a risk score and classify the threat level |
| **As a** developer, **I need** to validate system stability under multi-camera load conditions, **so that** the system's deployment readiness can be ensured | 5 | Enabler | -As a developer, if 4 cameras streams are active, then the system must remain stable, and performance logs must be recorded. |

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# APPENDIX

# 8. Appendix A :Interviews

*Table 7:Interview 1's Results - Release 1*

| Interview 1 | |
|---|---|
| **Question** | **Answer / Result** |
| 1. Name & Job Title | Laila – Security Supervisor / CCTV Monitor |
| 2. Years of Experience | 14 Years |
| 3. Work Environment | Security / Exhibition |
| 4. Average Number of Cameras Monitored Per Day | 12 to 16 cameras per day |
| 5. Type of Surveillance System Used | Traditional System |
| 6. How would you describe your overall experience with the current surveillance systems? | Exhausting; relies heavily on human focus, intuition, and speed; requires constant concentration. |
| 7. What is the most challenging part of your job when it comes to monitoring visitors? | Predicting behavior and distinguishing between normal and abnormal movements. |
| 8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage? | Never encountered; focus is purely on live monitoring |
| 9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds? | Yes, crowds cause distractions; a technical assistant (system) is necessary. |
| 10. How difficult is it to track a person of interest when they move between different camera views? | Tiring to monitor multiple cameras, especially if unorganized. |

| | |
|---|---|
| 11. What factors would make you trust and rely on an AI-based surveillance system? | Accuracy; not every human can maintain attention and recognize distractions constantly. |
| 12. During peak hours, how manageable is it for you to monitor all camera screens? | Cannot watch all screens at once; some areas might be missed. |
| 13. When a suspicious situation is identified, what are the usual steps and where do delays occur? | Team communication; delays occur if the receiver is not responsive. |
| 14. What type of alert would be most effective for you during work? | Sound notification combined with a visual alert. |
| 15.How important is it for you that the surveillance system operates locally without cloud transmissio | Prefers local storage (with visual cues) over cloud for security. |

*Table 8:Interview 2's Results - Release 1*

| Interview 2 | |
|---|---|
| **Question** | **Answer / Result** |
| 1. Name & Job Title | Abdulelah Qassem Al-Faifi – Perfume Salesperson |
| 2. Years of Experience | 6 Months |
| 3. Work Environment | Exhibition |
| 4. Average Number of Cameras Monitored Per Day | 6 cameras |
| 5. Type of Surveillance System Used | Traditional System |
| 6. How would you describe your overall experience with the current surveillance systems? | Primitive; depends fully on human monitoring without advanced technology. |
| 7. What is the most challenging part of your job when it comes to monitoring visitors? | High crowd density; difficult to focus and follow everyone simultaneously. |
| 8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage? | Yes, it happened more than once (detected after reviewing recordings). |

| | |
|---|---|
| 9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds? | Yes, human attention might miss fine details consistently. |
| 10. How difficult is it to track a person of interest when they move between different camera views? | Much harder as the number of cameras and customers increases. |
| 11. What factors would make you trust and rely on an AI-based surveillance system? | When the system is precise and accurate in detecting suspicious behaviors. |
| 12. During peak hours, how manageable is it for you to monitor all camera screens? | Very difficult; hard to maintain focus for a long period. |
| 13. When a suspicious situation is identified, what are the usual steps and where do delays occur? | Alerting the customer indirectly; delays occur when the person is at a distance. |
| 14. What type of alert would be most effective for you during work? | Visual alert on the dashboard. |
| 15.How important is it for you that the surveillance system operates locally without cloud transmissio | Very useful for small items to help make faster decisions. |

*Table 9:Interview 3's Results - Release 1*

| Interview 3 | |
|---|---|
| **Question** | **Answer / Result** |
| 1. Name & Job Title | Sultan – Self-Service Security Surveillance Supervisor |
| 2. Years of Experience | 18 Years |
| 3. Work Environment | Security / Exhibition |
| 4. Average Number of Cameras Monitored Per Day | 7 cameras |
| 5. Type of Surveillance System Used | Traditional System |
| 6. How would you describe your overall experience with the current surveillance systems? | Requires high concentration; reviewing recorded footage is very exhausting. |
| 7. What is the most challenging part of your job when it comes to monitoring visitors? | Monitoring during busy exhibition hours. |
| 8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage? | Yes, suspicious behavior was detected only after re-watching recorded footage. |
| 9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds? | Yes, must re-watch several times to detect small or slow movements. |

| | |
|---|---|
| 10. How difficult is it to track a person of interest when they move between different camera views? | Easy due to the small area, but the challenge is monitoring blind spots. |
| 11. What factors would make you trust and rely on an AI-based surveillance system? | Feels confident but believes there should always be a human backup. |
| 12. During peak hours, how manageable is it for you to monitor all camera screens? | Monitored daily; takes an entire day to review recordings properly. |
| 13. When a suspicious situation is identified, what are the usual steps and where do delays occur? | Reviewing recordings; delays occur due to camera blind spots. |
| 14. What type of alert would be most effective for you during work? | A mix of visual dashboard alerts and mobile notifications. |
| 15.How important is it for you that the surveillance system operates locally without cloud transmission | Yes, it would be helpful because it makes identifying events easier. |

*Table 10:Interview 4's Results - Release 1*

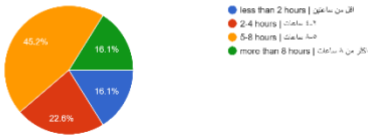
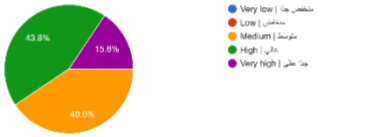
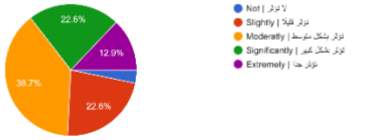
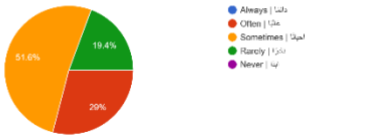
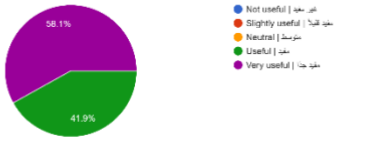
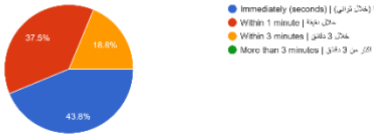
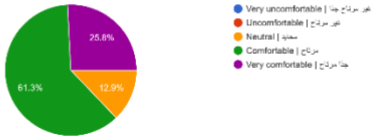
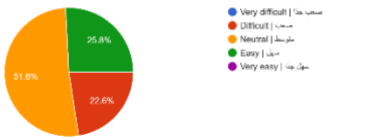
| Interview 4 | |
|---|---|
| **Question** | **Answer / Result** |
| 1. Name & Job Title | Eijan Al-Mandil – Staff |
| 2. Years of Experience | 9 Years |
| 3. Work Environment | Security / Exhibition |
| 4. Average Number of Cameras Monitored Per Day | 5 cameras |
| 5. Type of Surveillance System Used | Traditional System |
| 6. How would you describe your overall experience with the current surveillance systems? | Easy but not engaging or enjoyable. |
| 7. What is the most challenging part of your job when it comes to monitoring visitors? | Difficult camera angles. |
| 8. Have you ever encountered situations where suspicious behavior was only detected after reviewing recorded footage? | No, but equipment failures occur where the source of the fault is unknown. |
| 9. Do you believe that relying solely on human observation makes it difficult to detect small movements in crowds? | Hard to detect in unclear areas or when there is no audio. |
| 10. How difficult is it to track a person of interest when they move between different camera views? | Space is small, so no major difficulty except for blind spots. |
| 11. What factors would make you trust and rely on an AI-based surveillance system? | Trusts it for everything; useful for providing detailed reports on all activities. |

| | |
|---|---|
| 12. During peak hours, how manageable is it for you to monitor all camera screens? | Becoming easier now as it can be monitored via mobile. |
| 13. When a suspicious situation is identified, what are the usual steps and where do delays occur? | Delays occur due to internet issues or crowded spaces. |
| 14. What type of alert would be most effective for you during work? | Both sound and visual alerts together for better clarity. |
| 15.How important is it for you that the surveillance system operates locally without cloud transmissio | Prefers local storage over cloud for security. |

# 9. Appendix B: Questionnaires

Questionnaires Questions:

1. How many hours per shift do you spend monitoring cameras?
2. During monitoring, how high is your mental workload or focus level required?
3. How often do you feel that important events might be missed during live monitoring?
4. How quickly do you usually need to react when suspicious behavior is noticed?
5. How clear and easy to use is your current surveillance system interface?
6. How important is it for you to receive alerts with visual evidence (snapshot or short clip)?
7. How much do false alarms affect your work efficiency?
8. How useful would it be if the system prioritized the riskiest behavior automatically?
9. How comfortable are you with AI assisting you in monitoring tasks?
10. How important is system reliability (working without crashes or delays) during monitoring?
11. What features would help you perform your security tasks better?

| Q | Result | Q | Result |
|---|---|---|---|
| 1 | 1- How many hours per shift do you spend monitoring cameras? كم ساعة تقضي في مراقبة الكاميرات خلال الورديه الواحدة ؟<br>31 responses<br>  <br> | 6 | 6- How important is it for you to receive alerts with visual evidence (snapshot or short clip)? ما مدى اهمية استلام تنبيه يحتوي على دليل مرئي (صورة أو مقطع قصير)؟<br>31 responses<br> <br> |
| 2 | 2- During monitoring, how high is your mental workload or focus level required? اثناء المراقبة ، ما مستوى التركيز أو الجهد الذهني المطلوب منك ؟<br>32 responses<br>  <br> | 7 | 7- How much do false alarms affect your work efficiency? إلى أي مدى تؤثر الإنذارات الخاطئة على كفاءة عملك؟<br>31 responses<br>  <br> |
| 3 | 3- How often do you feel that important events might be missed during live monitoring? كم مرة تشعر انه قد يتم تفويت احداث مهمة اثناء المراقبة المباشرة ؟<br>31 responses<br>  <br> | 8 | 8- How useful would it be if the system prioritized the most risky behavior automatically? ما مدى فائدة ان يقوم النظام بإعطاء أولوية تلقائيا لأخطر السلوكيات؟<br>31 responses<br>  <br> |
| 4 | 4- How quickly do you usually need to react when suspicious behavior is noticed? ما السرعة المطلوبة عادة للاستجابه عند ملاحظة سلوك مشبوه ؟<br>32 responses<br>  <br> | 9 | 9- How comfortable are you with AI assisting you in monitoring tasks? ما مدى راحتك في استخدام نظام ذكاء اصطناعي لمساعدتك في المراقبة؟<br>31 responses<br>  <br> |
| 5 | 5- How clear and easy to use is your current surveillance system interface? ما مدى وضوح وسهولة استخدام واجهة نظام المراقبة الحالي؟<br>31 responses<br>  <br> | 10 | 10- How important is system reliability (working without crashes or delays) during monitoring? ما مدى أهمية موثوقية النظام (العمل بدون تعليق أو تأخير) أثناء المراقبة؟<br>31 responses<br> <br> |

| | |
|---|---|
| 11 | **11- What features would help you perform your security tasks better?**<br>**ما الميزات التي قد تساعدك على أداء مهامك الأمنية بشكل أفضل؟**<br>14 responses<br>Summarize responses<br>-Risk level prioritizations<br>-mukti camera suspect tracking<br>-snapshot evidence with alert<br>ان يكون هناك تنبية صوتي وأشعار عند حدوث شي مشبوه<br>- Clear location of suspensious activity<br>- Fewer false alarms<br>وجود سجل أحداث يمكن الرجوع له<br>التنبيه خلال وقت قصير جدا<br>- Silent alarms on dashboard<br>-Record automatic snapshots for future use and documentation.<br>-System that works fully offline<br>-behavior based alerts ( not just motion)<br>وجود منبة<br>-A "false alarm feedback" button so i can train the system<br>يكون هناك نظام مساعد متطور يساعد الفرد فيه التنبيه عند وجود مشتبه وايضا يتتبع المشتبه به لان هناك صعوبه بتتبع جميع الشاشات والكاميرات<br>١-تحديد موقع الحدث<br>٢-واجهة سهلة وواضحة<br>-Color coded risk level to help me react faster<br>-One notification instead of multiple for the same behavior<br>يكون هناك مساعد ويتتبع تحركات المشتبة به<br>ان يعمل بدون انترنت، امكانية إرجاع الفيديو |

**Form's link:** https://forms.gle/sD988Lv4VBejDv5aA

# 10. Appendix C: GitHub

**GitHub link:** https://github.com/lanaabdullah1/2025-GP1-3.git

# 11. Appendix D: Jira

**Jira link:** https://joudkja1.atlassian.net/jira/software/projects/DK2025/boards/3